ELMASNUR YILMAZ

PHD CANDIDATE

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EDUCATION

Ph.D.

Ege University
Graduate School of Natural and Applied Sciences
Department of Biotechnology September 2020 – cont.
Thesis Proposal Title: "Investigation of TDP-43 Mediated Regulation of Cryptic Exons and Identification of New Candidate Genes"
Advisor: Prof. Dr. Yasemin ERAÇ

Master's Degree

Dokuz Eylul University
Izmir Biomedicine and Genome Institute December 2018
Molecular Biology and Genetics
Thesis: "New genomic approaches to explore the neurogenetic disease burden of consanguineous marriages in Turkey"
Advisor: Dr. Yavuz OKTAY

Bachelor's Degree

Istanbul Technical University February 2016
Molecular Biology and Genetics
Thesis: "The usage of salt leaching method in the construction of porous tissue culture scaffolds"
Advisor: Prof. Dr. Fatma Neşe KÖK

SCHOLARSHIPS

BİDEB (The Scientific and Technological Research Council of Türkiye) 2211-A National PhD Scholarship Program
September 2021– August 2025

YÖK (Council of Higher Education of Türkiye) 100/2000 Molecular Pharmacology and Pharmaceutical Research Fellow September 2020 – August 2024

TUBITAK Project Scholar October 2017 – December 2018
"New Genomic Approaches to Explore the Neurogenetic Disease Burden of Consanguineous Marriages in Turkey"

TUBITAK Project Scholar November 2016 – September 2017
"Uncovering the Molecular Mechanisms of the SNP rs55705857 in the 8q24.21 Region Predisposing to Gliomas"

PROJECTS

Researcher TUBITAK October 2024 – October 2025
Project Number: 124S480
"Effects of *Morus nigra L., Salvia tomentosa* Mill. and *Salvia dichroantha* Stapf. Extracts on Virulence Factors of *Stenotrophomonas maltophilia* and *Acinetobacter baumannii* Strains"

Researcher Ege University BAP September 2023-September 2025
Project Number: 31901
"Investigation of TDP-43 Mediated Regulation of Cryptic Exons and Identification of New Candidate Genes"

Researcher Ege University BAP May 2023- May 2024
Project Number: 29657
"Investigation of Anticancer Effect and Mechanism of Extracts from *Salvia tomentosa* Above Ground Parts on Different Cancer Cell Lines"

Researcher TUBITAK October 2017 – December 2018
Project Number: 216S771
"New Genomic Approaches to Explore the Neurogenetic Disease Burden of Consanguineous Marriages in Turkey"

Researcher TUBITAK November 2016 – September 2017
Project Number: 214S097
"Uncovering the Molecular Mechanisms of the SNP rs55705857 in the 8q24.21 Region Predisposing to Gliomas"

EXPERIENCE

Ege University, İzmir, Türkiye Sep 2020/cont.
PhD Student, Eraç Lab, Faculty of Pharmacy, Department of Pharmacology
Involved in both wet-lab and computational research with a focus on transcriptomic alterations and functional pharmacological assays. Responsibilities included routine cell culture studies, PCR and qPCR experiments, viability assays (MTT, WST), calcium imaging experiments, and pharmacological drug response assays.

Additionally, carried out comprehensive bioinformatics analysis using RNA-seq data: retrieval of raw FASTQ files, quality control, alignment to reference genomes, alternative splicing and cryptic exon detection (rMATS), and downstream statistical and visual analysis using R (DESeq2, ggplot2) and Python.

Newcastle University, Newcastle upon Tyne, United Kingdom Jun 2018/Jul 2018
Visiting Student, Institute of Genetic Medicine/International Centre for Life
Whole exome sequencing data analysis, fibroblast cell culture, western blotting, neural progenitor cell studies

Dokuz Eylul University, İzmir, Türkiye Feb 2016/Dec 2018
Researcher, Oktay Lab, Izmir Biomedicine and Genome Center
Collection of blood samples from patients, isolation of DNA and RNA from samples, preliminary screening with routine PCR and gel electrophoresis, Sanger sequence analysis, whole exome sequencing data analysis, routine cell culture studies and analysis, CRISPR-Cas studies

Istanbul Technical University, İstanbul, Türkiye Feb 2015/June 2015
Undergraduate Researcher, Kök Lab
"The usage of salt leaching method in the construction of porous tissue culture scaffolds"
Basic molecular methods in tissue engineering laboratory for undergraduate thesis

TUBİTAK/MAM, Kocaeli, Türkiye Jun 2014/Jul 2014
Intern, Genetic Engineering and Biotechnology Institute
"Production of Monoclonal Antibody Using Hybridoma Technology"
Cell culture studies (passage, counting, freezing), indirect ELISA

Marmara University, İstanbul, Türkiye Aug 2013/Sep 2013
Intern, Bioengineering
"Investigation of Beta-Lactamase Allosteric Site Ligand Recognition and Binding"
Primer design, SDS-PAGE, sonication, and enzyme activity assays

Royal Institute of Technology (KTH), Stockholm, Sweden Jun 2011
Participant
"S(t)imulating life: from CPU to your stomach"
Protein simulation using GROMACS

PUBLICATIONS

Demir, S., Yasar, B. Y., **Yilmaz**, **E**., Karasu, A. Z., Erac, Y., & Baykan, S. (2025). Investigation of Cytotoxic Activity of Salvia tomentosa and Methyl Carnosate-12-Methylether on Different Cancer Cell Lines. ***Revista Brasileira de Farmacognosia***, 1-11.

Kurul, S. H., Oktay, Y., Töpf, A., Szabó, N. Z., Güngör, S., Yaramis, A., ... **Yılmaz, E**., … & Horvath, R. (2021). High diagnostic rate of trio exome sequencing in consanguineous families with neurogenetic diseases. ***Brain***.

Gungor, S., Oktay, Y., Hiz, S., Aranguren-Ibáñez, Á., Kalafatcilar, I., Yaramis, A., ... **Yılmaz, E**., … & Horvath, R. (2021). Autosomal recessive variants in TUBGCP2 alter the γ-tubulin ring complex leading to neurodevelopmental disease. ***Iscience**,* 24(1), 101948.

Oktay, Y., Gungor, S., Hiz, S., Yaramis, A., Aranguren-Ibanez, A., Yis, U., ... **Yılmaz, E**., … & Horvath, R. (2020). A homozygous missense variant inTUBGCP2alter the g-tubulin ring complex leading to abnormal cortical development, pontocerebellar atrophy and altered myelination. ***European Journal Of Human Genetics***, 28(Suppl 1), 460–461.

Yaramis, A., Lochmüller, H., Töpf, A., Sonmezler, E., **Yilmaz, E**., Hiz, S., ... & Beltran, S. (2020). COL4A1-related autosomal recessive encephalopathy in 2 Turkish children. ***Neurology Genetics***, 6(1).

Oktay, Y., Güngör, S., Zeltner, L., Wiethoff, S., Schöls, L., **Yılmaz, E**., ... & Kaemereit, S. (2020). Confirmation of TACO1 as a Leigh Syndrome Disease Gene in Two Additional Families. ***Journal of neuromuscular diseases,*** 7(3), 301-308.

Topf, A., Oktay Y., Balaraju S., **Yilmaz E**.,& Hiz S., Horvath, R. (2019). Severe neurodevelopmental disease caused by a homozygous TLK2 variant. ***European Journal of Human Genetics***. 2019 June. doi:10.17863/CAM.40761.

Yilmaz, E., Sonmezler, E., Topf, A., Balaraju, S., Yaramis, A., Gungor, S., ... & Hiz, S. (2019, July). A novel mutation in the coiled-coil interaction domain of LAMB1 extends the molecular basis of laminin-related cortical malformation phenotypes. In ***European Journal of Human Genetics*** (Vol. 27, pp. 285-286).

POSTERS

Elmasnur Yilmaz, Yasemin Erac (2024, July). "Tdp-43 Mediated Splicing and Its Intersection with Ubiquitination In Gene Regulation" 3rd Ubiquitin Function in Health & Disease Conference 07 – 10 July 2024 Lisboa, Portugal.

Elmasnur Yilmaz, Beste Yurdacan Yasar, Damla Getboga, Turkum Dilan Sen, Ilay Nur Karstarli, Yasemin Erac (2024, June). "The Effects of Hyperforin in the Sorafenib-Resistance Hepatocellular Carcinoma" EPHAR 2024 The Federation of European Pharmacological Societies 9th European Congress of Pharmacology 23-26 June 2024 Athens, Greece.

Elmasnur Yilmaz, Lingtao Peng, Melissa J Moore, Yasin Kaymaz (2023, Sep). "Disruption of TDP43 expression leads to transcriptome-wide deregulation of polyadenylation sites" ICGEB Arturo Falaschi Conference "1st biennial conference on TDP-43 function and dysfunction in disease" 06/09/2023 to 08/09/2023 Trieste, Italy.

Hız, S.,**Yılmaz, E**.,& Oktay, Y.(2018, Nov). CONSEQUITUR: A multinational effort towards determining the neurogenetic disease burden of consanguineous marriages in Turkey by genomics approaches. Poster presented at the 15th International Child Neurology Congress (ICNC 2018), Mumbai, India.

Gungor, S., Okur, D., **Yılmaz, E**.,& Hız, S. (2018,Nov). Two distinct phenotypes caused by the same mutation in the SAMHD1 gene. Poster presented at the 15th International Child Neurology Congress (ICNC 2018), Mumbai, India.

Topf,A., Oktay, Y., Balaraju, S., **Yılmaz, E**., & Horvath, R. (2018). MITOCHONDRIAL DISEASES (Posters): P.193 Unexpected genetic diagnosis of mitochondrial disease in three consanguineous Turkish families. Neuromuscular Disorders, 28,s85-s86.

TALKS

Elmasnur YILMAZ (2023, Sep). "Disruption of TDP43 expression leads to transcriptome-wide deregulation of polyadenylation sites" ICGEB Arturo Falaschi Conference "1st biennial conference on TDP-43 function and dysfunction in disease" Trieste, Italy, 06/09/2023 to 08/09/2023.

Elmasnur YILMAZ (2018, Sep). "Better Call RoH: What a 11Mb Homozygous Haplotype Block Can Tell Us About a Possible Founder Mutation in the Turkish Population" 6th International Congress of the Molecular Biology Association of Turkey (MolBiyoKon Conference 2018), Izmir, Turkey.

Elmasnur YILMAZ (2018, Dec). Moleküler Genetikteki Son Gelişmelerin Çocuk Nöroloji Hastalıklarının Tanı ve Tedavisine Katkıları: RD-Connect Platform Aracılığıyla Sekans Analizi. (Contributions of Latest Developments in Molecular Genetics to the Diagnosis and Treatment of Child Neurology Diseases: Sequence Analysis via RD-Connect Platform). Neurology Seminar Izmir, Turkey

LANGUAGE

English: CEFR Level C1 IELTS Test Result:7 (Listening: 7.0 Reading: 8.0 Writing: 6.0 Speaking: 6.0 Test Date: January 2018)

YÖKDİL English Exam Result: 96.25/100 (Exam Date: 13.03.2022)

PROFESSIONAL MEMBERSHIPS

- European Association for Cancer Research (EACR) Membership Number: EACR35041
- The Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment (ISTAART) Membership Number: 100020374703
- Istanbul Technical University Alumni Association (Mentor)

CERTIFICATES

- The Computational Genomics Course 2025 (Compgen2025) Module 1. The module focused on AI-assisted data analysis and was held between 3rd March and 9th March 2025
- The Computational Genomics Course 2025 (Compgen2025) Module 3.The module focused on "Multi-omics Data Integration Using Deep Learning Approaches With Applications In Precision Oncology" and was held between 10th-16th March 2025.
- "Fundamentals of AI/ML in Precision Medicine" from Stanford Deep Data Research Center April 18, 2024
- "Fundamentals of Data Science in Precision Medicine and Cloud Computing" from Stanford Deep Data Research Center April 07, 2024

BIOINFORMATICS & COMPUTATIONAL BIOLOGY SKILLS

- **High-throughput Sequencing Data Analysis (RNA-seq)**: Proficient in preprocessing and quality control (*FastQC, MultiQC*), adapter trimming (*Trim Galore, Cutadapt*), alignment to reference genomes (*HISAT2, STAR*), and post-alignment processing (*SAMtools, BAMtools, Picard*).
- **Alternative Splicing & Cryptic Exon Discovery**: Hands-on experience in identifying differential splicing and cryptic splicing events using *rMATS* and *MAJIQ*, including visualization of splice junctions and exon usage patterns.
- **Transcriptome Assembly & Quantification**: Familiar with transcript assembly (*StringTie*) and read quantification using *featureCounts* and *HTSeq-count*; proficient in genome annotation parsing with *GENCODE*/*Ensembl* GTF/GFF files.
- **Pipeline Development & Workflow Automation**: Able to design and execute reproducible pipelines using *Shell scripting*, *Snakemake*, and *Nextflow* in Unix-based environments; experienced in automating large-scale data tasks efficiently.
- **Data Retrieval & Management**: Skilled in fetching and handling public datasets (e.g., *ENA/SRA* using prefetch, fasterq-dump), converting and organizing *FASTQ/BAM* files, and maintaining file integrity with indexing tools.
- **Statistical & Differential Expression Analysis in R**: Proficient in using *DESeq2*, *edgeR*, and *limma*; experienced in exploratory data analysis such as *PCA*, clustering, and pathway enrichment (*clusterProfiler*, *EnhancedVolcano*, *ComplexHeatmap*).
- **Data Visualization**: Advanced data visualization in **R** (*ggplot2, tidyverse, plotly*) and **Python** (*matplotlib, seaborn, pandas, plotnine*); generation of publication-quality plots for complex datasets.
- **Genome Annotation & Coordinate Manipulation**: Experience with *BEDTools*, *GTFtools*, *GffCompare*, and strong understanding of genomic coordinate systems (*BED, GTF, VCF*).
- **Programming Languages**:
  - **Python**: Data wrangling, automation, and visualization using *Pandas*, *Seaborn*, *Matplotlib*, and Jupyter Notebooks (primarily in *VSCode*).
  - **R**: Statistical genomics workflows using *RStudio* with *DESeq2*, *ggplot2*, *ComplexHeatmap*.
  - **Shell scripting**: Used for workflow automation and batch processing on Unix/Linux systems.
- **Command-line Proficiency**: Fluent in Linux/Unix terminal environments for scripting, bioinformatics tool integration, and file system management.

- **Environment & Package Management**: Comfortable working with *Conda* and *Mamba* environments; experienced in troubleshooting tool installations on Mac M1 architecture.
- **Version Control & Reproducibility**: Proficient with *Git*/*GitHub* for version control, collaboration, and reproducible research documentation.
- **Cloud & HPC Computing**: [e.g., Google Colab, AWS, institutional HPC clusters.]
- **Soft Skills**:
    - Strong scientific curiosity and a problem-solving mindset in computational biology.
    - Capable of communicating complex data through clean and informative visualizations.
    - Interdisciplinary collaboration experience bridging wet-lab biology with computational analysis.